

1. DIGIT COUNT:

```
public class Main1 {  
    public static void main(String[] args) {  
        int num=12345;  
        int count=0;  
        while(num>0){  
            count++;  
            num=num/10;  
        }  
        System.out.println(count);  
    }  
}
```

2. Backward digit iteration:

Sample input	Sample output
12345	5 4 3 2 1

solution:

```
public class Main2 {  
    public static void main(String[] args) {  
        int num=12345;  
        while(num>0){  
            int digit= num%10; //1234%10=4  
            System.out.print(digit+ " "); //5 4  
            num=num/10; // 123  
        }  
    }  
}
```

3. Forward digit iteration:

Sample input	Sample output
12345	1 2 3 4 5

Solution:

```
public class Main3 {
    public static void main(String[] args) {
        int num=12345;
        int temp=num;
        int count=0;
        while(temp>0){

            temp=temp/10;
            count++;
        }
        System.out.println(num+" "+count);
        int div=1;
        for(int i=1;i<count;i++){ // finding divisor without using math.pow
            div=div*10;
        }
        System.out.println(div);

        while(div>0){
            int digit=num/div;//12345/10000=1
            num=num%div;//updated num=2345
            System.out.print(digit+" ");//1
            div=div/10;//1000
        }
    }
}
```

NESTED LOOP TRACING REVIEW:

<https://drive.google.com/file/d/12lb1m-IMbclJDplmo6vDJoTxusteuZKj/view?usp=sharing>

Practice SUGGESTIONS:

SOLVE ALL THE HOME TASK, CLASS TASK, UNGRADED TASK FROM LAB ASSIGNMENT 1 TO 5 FOR PRACTICE.